Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · DISTRIBUTION & NORMALITY

is a variable normally distributed?

Table. Normality tests

Variable	Test	Statistic	N	p	Skew	Kurtosis (excess)
exam_score	Shapiro-Wilk	W 0.99	240	.158	−0.18	−0.53
exam_score	K–S (Lilliefors)	D 0.04	240	.608	−0.18	−0.53
absences	Shapiro-Wilk	W 0.86	240	<.001	1.50	2.43
absences	K–S (Lilliefors)	D 0.17	240	<.001	1.50	2.43

Kurtosis is excess (normal = 0); both via psych::describe (type 3). Shapiro-Wilk applies for 3–5000 cases.

Figure. Distribution shape - exam_score

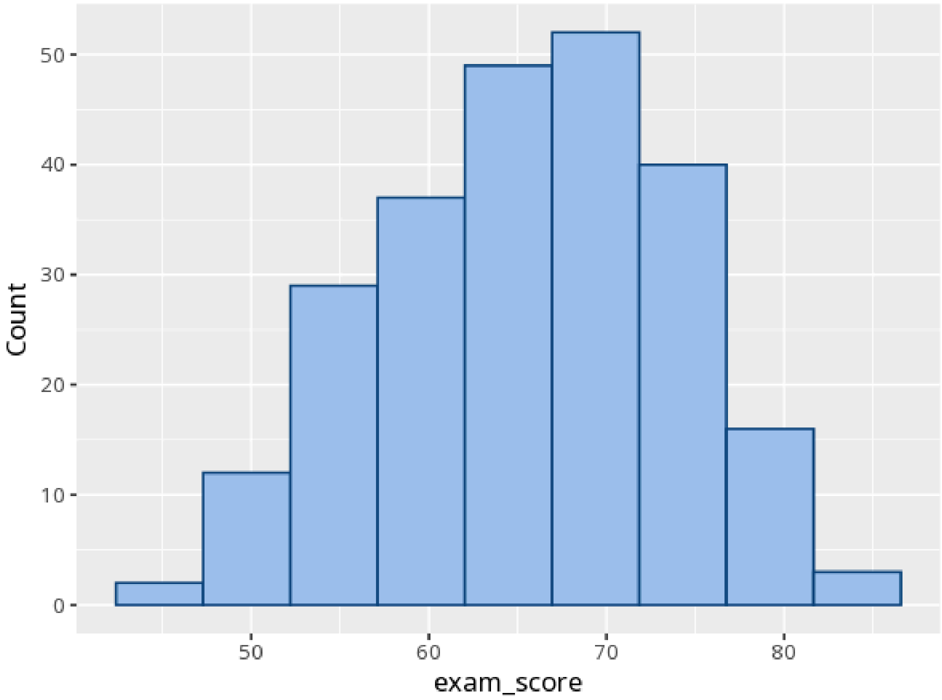


Figure. Distribution shape - exam_score

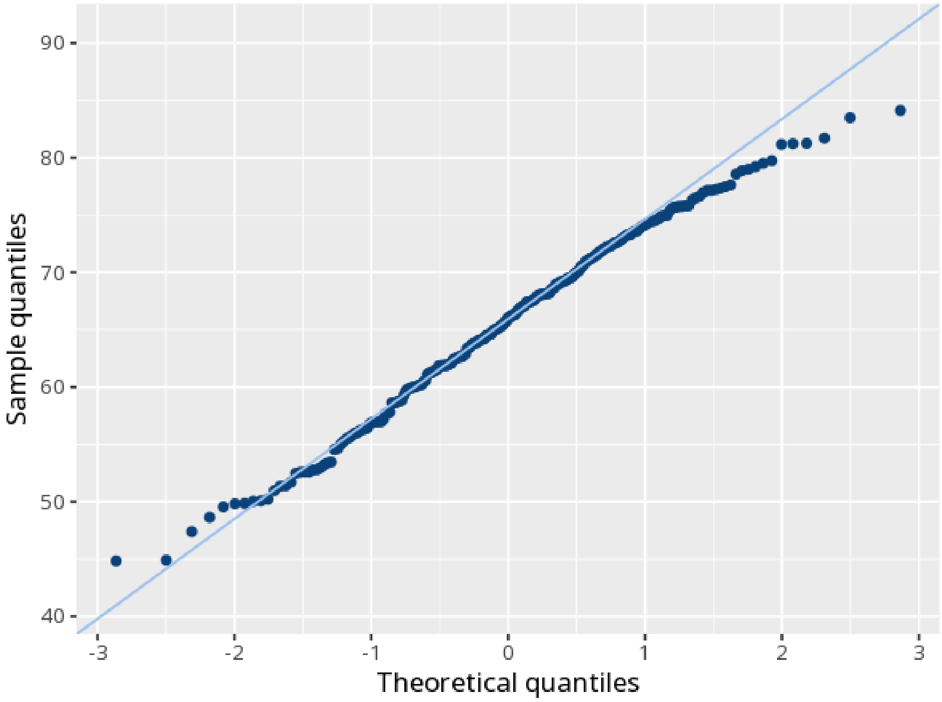


Figure. Distribution shape - absences

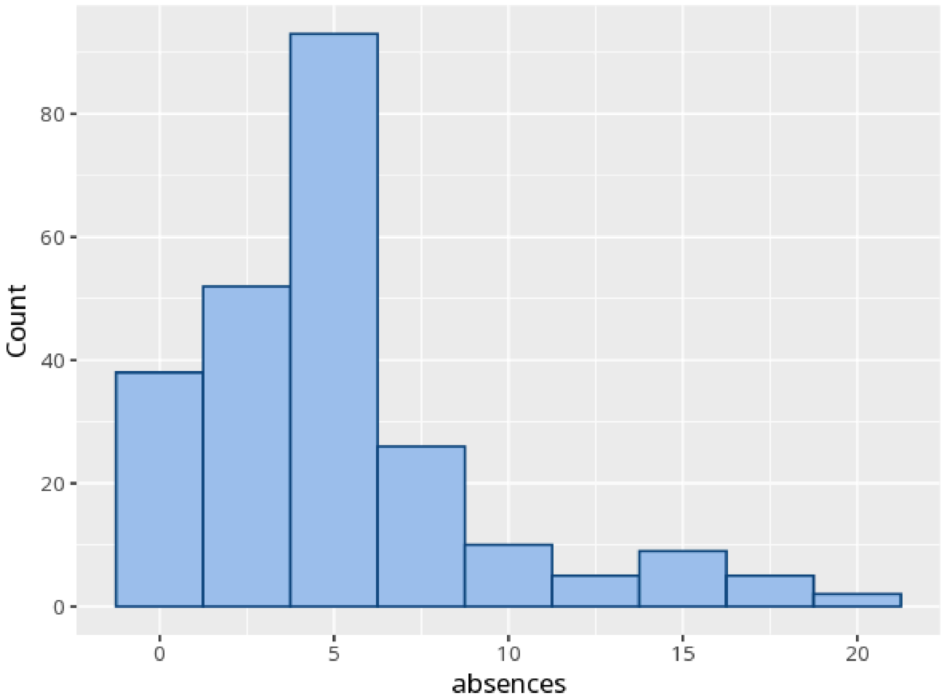
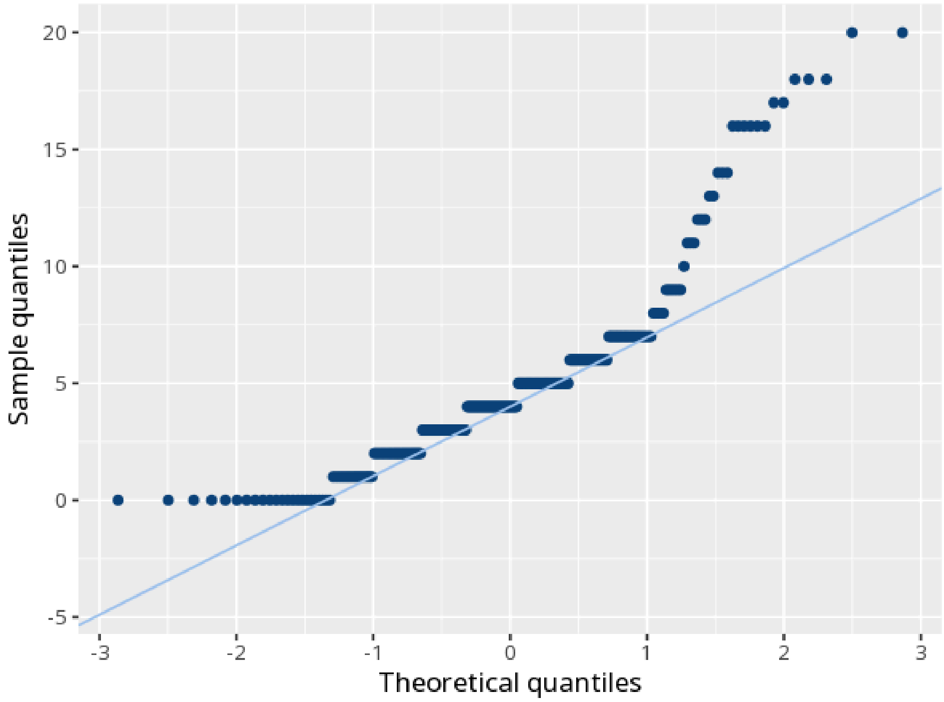


Figure. Distribution shape - absences



Understanding the numbers

Variable. Which variable that row's normality test describes. This run tests normality for 2 variable(s).

Test. Which normality test that row reports - Shapiro-Wilk or K-S (Lilliefors) - both computed so you can cross-check them. Here, both tests are run for each of the 2 variable(s), but judge normality mainly from the Q-Q plot - points hugging the diagonal indicate normality.

Statistic (W / D). The test statistic itself - W for Shapiro-Wilk, D for K-S (Lilliefors) - summarizing how far the sample departs from a normal distribution. Here, 2 of 4 computed tests flag a significant departure from normality ($p < .05$).

N. How many valid values that variable's test was computed on (Shapiro-Wilk needs 3-5000 cases; K-S needs at least 5). Here, N = 240 across the reported variable(s).

p. The probability of a departure from normality this large (or larger) if the data truly were normal; with large samples this over-flags trivial deviations, and with small samples it has low power. Here, 2 of 4 computed tests are significant ($p < .05$) - either way, check the Q-Q plot before concluding.

Skew. How far the distribution's shape departs from a normal bell curve in asymmetry. Here, Skew = −0.18–1.50 across the reported variable(s).

Kurtosis (excess). How far the distribution's shape departs from a normal bell curve in tail weight (normal = 0). Here, Kurtosis = −0.53–2.43 across the reported variable(s).

How to read this test

A small p (below .05) suggests the data depart from normality. With large samples these tests over-flag trivial deviations; with small samples they have low power, so a $p > .05$ does not prove normality - it only means no departure was detected. Either way, judge normality mainly from the Q–Q plot - points hugging the diagonal indicate normality.

APA template: exam_score: Normality was assessed with the Shapiro-Wilk test, W = .99, p = .158.
absences: Normality was assessed with the Shapiro-Wilk test, W = .86, p < .001.

Statistical basis: Shapiro-Wilk test for normality (3-5000 cases) (Shapiro, S. S., Wilk, M. B. (1965). "An analysis of variance test for normality (complete samples)." Biometrika, 52(3/4), 591-611.); Additional normality tests (Lilliefors and related) (Gross J, Ligges U (2015). nortest: Tests for Normality. R package.). Full references in the exported CITATIONS.txt.